



NovoRetin™
The natural alternative
to retinol



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A Precious Medicinal Resin to Boost Retinoic Acid Effects in the Skin

Based on mastic, which is an aromatic resin from the bark of a tree that grows on the Greek island of Chios, NovoRetin™ is able to increase the amount of naturally occurring retinoic acid in the skin, which leads to retinol-like effects.

Retinol is well-known for its strong anti-aging effects and the benefits it offers for acne-prone skin. Retinoic acid, which is the major active metabolite of retinol, is responsible for these retinol effects in the skin. NovoRetin™ is able to inhibit the activity of CYP26 enzymes that degrade retinoic acid in the skin. Thereby, NovoRetin™ can retain naturally occurring retinoic acid levels in the skin, which leads to retinol-like effects without the need to apply retinoids topically on the skin.

Clinical studies conducted on Caucasian and Asian women have shown that 2 % NovoRetin™:

- increases skin elasticity and skin density by 20 % and 14 % after 28 days
- has an immediate lifting potential with an improvement in wrinkle depth and length by 14 % after only a single application
- refines pores and reduces pore volume by 42 % after only two weeks
- reduces retentional lesions with a reduction in black-heads by 40 %
- reduces shininess by 11 % after only two weeks, while increasing skin hydration at the same time.

Therefore, NovoRetin™ serves as an ideal plant-based retinol alternative in cosmetics, providing both strong anti-aging effects and exceptional benefits for acne-prone skin.

NovoRetin™

- Boosts naturally occurring retinoic acid in the skin
- Improves skin elasticity and density
- Effects immediate lifting
- Visibly refines pores and blemishes
- Stable and safe retinol activity

Applications

- Advanced age-defying formulas
- Immediate lifting products
- Treatments for acne-prone skin
- Pore minimizing formulas

Formulating with NovoRetin™

- Recommended use level: 2–3 %
 - Incorporation: For cold processes, dissolve NovoRetin™ into the aqueous phase. In hot/cold processes, add during the cooling phase below 40°C.
 - Thermostability: Temperatures of up to 40°C for a short time will not affect the stability of NovoRetin™.
- For further information, please refer to our formulation guidelines.

INCI (EU/PCPC) Declaration

Pistacia Lentiscus Gum / Pistacia Lentiscus (Mastic) Gum (and) Lecithin (and) Pentylene Glycol (and) Glyceryl Caprylate / Caprate (and) Caprylic / Capric Triglyceride (and) Aqua / Water

Additional Information

- Liquid
- Water-soluble
- Without preservatives
- 100 % natural origin

Mastic

An endemic resin from the Greek island of Chios

Mastic, the Gold of Chios since Ancient Times

Chios mastiha, which is also known as mastic, is an aromatic resin that is produced by the lentisk tree, and it grows exclusively on the Greek island of Chios (*Pistacia lentiscus* var. Chia). The name is derived from the Greek words "μαστάζω" (chew) and "μάσταξ" (mouth). Mastic is excreted in the form of tears from the trunk and the main branches after scratches are made on their surfaces. After its solidification, which results in an ivory translucent colored resin, it is collected, washed, and cleaned. This traditional and sustainable harvesting process makes mastic a renewable resource that supports the communities on the Greek island of Chios.

Mastic has been recognized since ancient times for its unique aroma and therapeutic properties. However, due to its high price in ancient times mastic was only purchased by wealthy people.

Mastic Trees only Weep in Chios

The mastic tree or lentisk is an evergreen shrub that grows on arid, rocky, and poor soil. Lentisks are part of the typical maquis-type vegetation that is found in Mediterranean countries.

However, only the mastic trees of southern Chios give the mastiha resin when their bark is scratched. This is probably due to the soil and especially warm and dry weather conditions. And it explains why mastic production in Chios is granted protected designation of origin (PDO). The know-how relating to the cultivation of mastic on the island of Chios was also inscribed on the "list of the intangible cultural heritage of humanity" by UNESCO in 2014.

Gentle and Sustainable Harvesting of the Mastic Tears



Mastic

An old known beauty remedy

Medicinal, Pharmaceutical, and Industrial Applications

Mastic was the first natural chewing gum in the ancient world. It was used to clean teeth and freshen breath, as well as prevent and cure digestive disorders. Scientific studies have confirmed these historically recorded effects: chewing mastic improves oral hygiene by reducing microbial plaque formation and eliminating bacterial growth inside the oral cavity. Furthermore, chewing mastic contributes to the soothing of the stomach.

Due to its anti-bacterial, anti-fungal, anti-inflammatory, and antioxidant effects, mastic has also been used in recent years:

- in the preparation of ointments for skin disorders as well as in toothpaste and mouthwash for cleaning and disinfecting the oral cavity
- as a component of bandages, adhesive plasters, and compresses for the protection and healing of wounds, as well as for dental fillings and tooth molds.

Mastic for Skin Care

Mastic has been consistently used in skin care formulations for hundreds of years.

In the 1st century AD, Dioscorides, a renowned pharmacologist, wrote that mastic promoted skin radiance when it was incorporated into face creams. In addition, he noted that mastic oil cleansed the face and gave it a healthy color.

Meanwhile, in the Roman Empire, upper-class ladies used beauty creams that contained mastic, while in the Byzantine Empire mastic was used to produce luxury scented soaps.

NovoRetin™

The application of mastic in modern cosmetics, however, has been limited due to its stickiness and insolubility in water. This is no longer an issue as Mibelle Biochemistry has developed a unique delivery system to make mastic bioavailable for the skin. The resulting water-soluble product NovoRetin™ is easy to formulate and it offers an innovative alternative to retinol in skin formulations, as it enhances the levels of naturally occurring retinoids in the skin.

Collecting the Resin and Cleaned Mastic Tears



Retinol Effects on the Skin

Retinoic acid boosting by CYP26 inhibition

Use of Retinoids for Skin Applications

Retinol (vitamin A) and structurally related molecules of the group of retinoids are required for many biological processes in the human body. Therefore, they occur naturally in human organs including the skin. Due to their modulatory function on keratinocyte proliferation, retinoids can influence the maintenance of the skin and have positive effects on wound healing and skin barrier function. However, retinoids are also well-known in the cosmetic field due to their strong anti-aging properties when being applied topically. Furthermore, retinoids are used to treat pathological skin conditions, including acne, psoriasis, and ichthyosis.

Retinoic Acid

Retinoic acid is the major active metabolite of retinol. It exists in several isoforms, the most common of which are all-trans retinoic acid and 9-cis retinoic acid. The activity of retinoids is mediated mainly through binding of retinoic acid to specific nuclear receptors and thereby regulating gene expression. Like other retinoids, retinoic acid is not very stable in formulations and the treatment with high concentrations of retinoic acid can lead to severe adverse effects, limiting its chronic therapeutic application and completely preventing its use in cosmetics.

CYP26 Enzymes Degrade Retinoic Acid

Retinoic acid is able to rapidly induce its own degradation by upregulating the expression of cytochrome P450 family 26 (CYP26) enzymes, which degrade retinoic acid into hydroxy-retinoic acid. In the skin, CYP26A1, which is a member of this enzyme family, is responsible for retinoic acid degradation, as demonstrated in cultured keratinocytes (1).

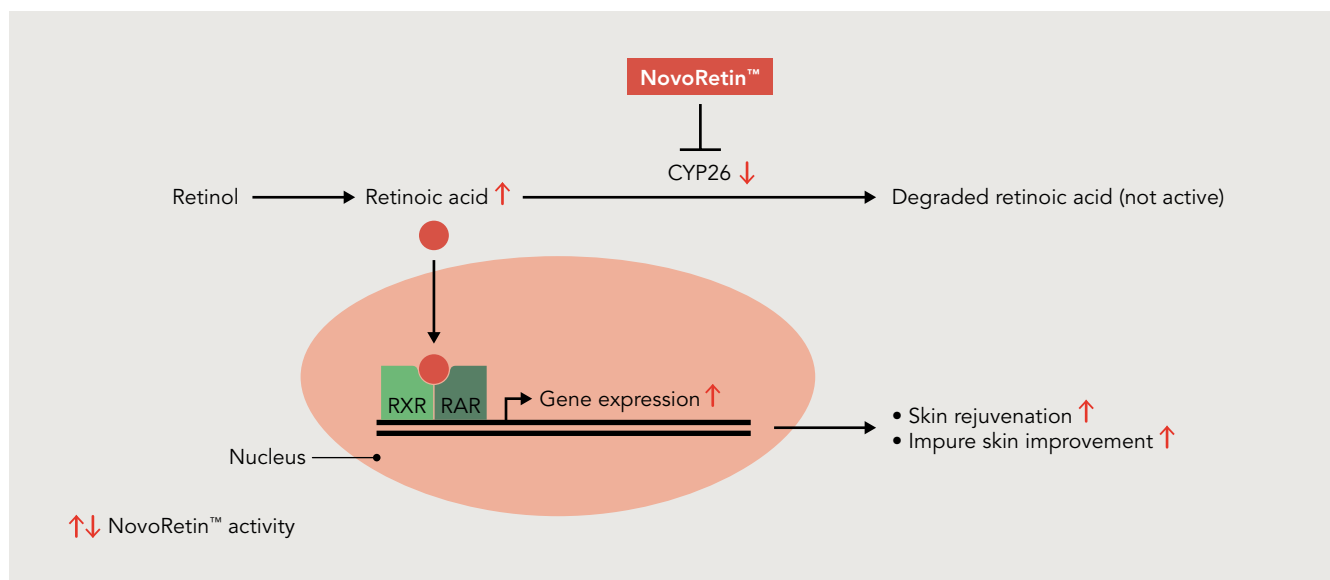
Boosting of Retinoic Acid Effects by CYP26A1 Inhibition in the Skin

Due to the induction of CYP26 enzymes, retinoic acid is degraded quickly and therapies with topically applied retinoids require high concentrations. A strategy to overcome these limitations is to increase the levels of naturally occurring retinoic acid within the skin by inhibiting its degradation. Specific CYP26 inhibitors, such as liarozole or talarozole, have demonstrated efficacy in several dermatological disorders with better tolerability than previous retinoid therapies (2). NovoRetin™ is a natural, plant-based alternative for cosmetic applications to inhibit CYP26A1 activity in the skin and thereby increase retinoic acid activity.

(1) Heise, R. et al. (2006) *J Invest Dermatol* 126(11): 2473–80.

(2) Verfaillie, C. J. et al. (2008) *J Dtsch Dermatol Ges* 6(5): 355–64.

A Mechanism to Increase Endogenous Retinoic Acid Levels



Natural Retinoic Acid Booster

Anti-aging effects & benefits for impure skin

Natural Retinol Alternative for Cosmetics

NovoRetin™ is a natural CYP26A1 inhibitor that can prevent the degradation of retinoic acid in the skin. Therefore, by using NovoRetin™, naturally occurring retinoic acid in the skin can be retained, which leads to retinol-like effects in the skin without the need to apply retinoids topically on the skin. Using NovoRetin™, formulation challenges due to the instability of retinoids and the negative side effects of topical retinoid treatments are completely avoided.

Strong Anti-Aging Effects

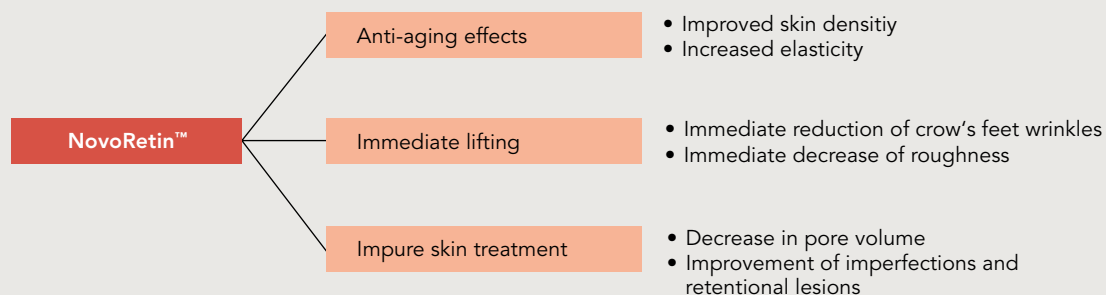
By boosting retinoic acid levels in the skin, NovoRetin™ demonstrates well-known retinol-like anti-aging effects, such as an improvement in skin elasticity and density. An additional benefit of NovoRetin™ and indeed an advantage compared to retinoids is its immediate anti-wrinkle potential and its ability to minimize pores that enlarge during skin aging.

Benefits for Impure Skin

By enhancing retinoic acid in the skin, NovoRetin™ also has positive effects on impure and acne-prone skin. Clinical studies have demonstrated its anti-comedogenic effect and its ability to reduce shininess and skin imperfections. In addition, the efficacy of NovoRetin™ to minimize pores is also a great benefit for this skin type.

Therefore, based on an innovative mechanism, NovoRetin™ serves as a plant-based retinol alternative in cosmetics that can be used for both anti-aging applications and formulations for acne-prone skin. This means it is also an ideal active ingredient for blemished skin showing first signs of skin aging.

Activities of NovoRetin™



NovoRetin™

Study results



Inhibition of Retinoic Acid Degrading Enzyme Activity

The effect of NovoRetin™ on the inhibition of CYP26A1 was investigated by a metabolic activity assay in human keratinocytes (HaCaT cells). For this assay, cells were treated with NovoRetin™ at different concentrations combined with radiolabeled all-trans retinoic acid for 9 hours. Following this, total metabolites were extracted from the cells, and metabolites of the labeled retinoic acid were quantified with a liquid scintillation counter to determine CYP26A1 activity. Treatment of keratinocytes with NovoRetin™ decreased CYP26A1 activity dose-dependently, demonstrating the potential of NovoRetin™ to inhibit CYP26A1 activity in the skin.



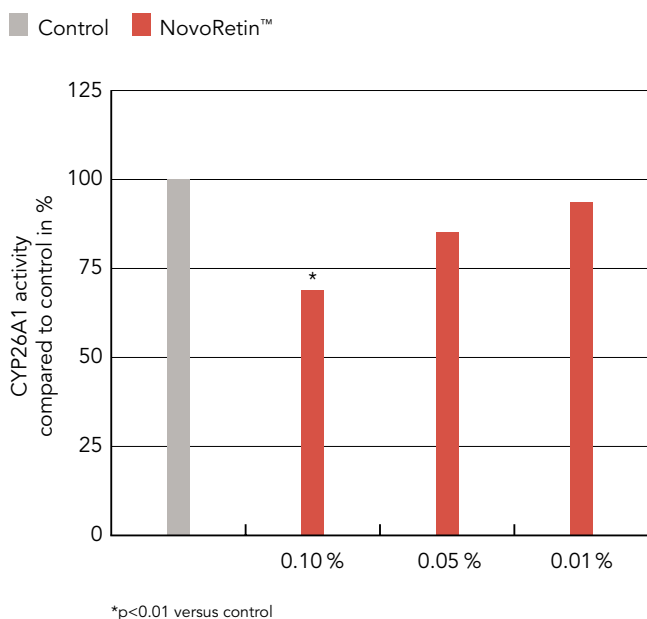
Inhibition of Retinoic Acid Degrading Enzyme Expression

Inhibition of CYP26A1 was further tested on gene expression level. Keratinocytes (HaCaT cells) were incubated with all-trans retinoic acid in the presence of NovoRetin™ at different concentrations for 4 hours. RNA was extracted and the CYP26A1 gene expression was determined by RT-qPCR.

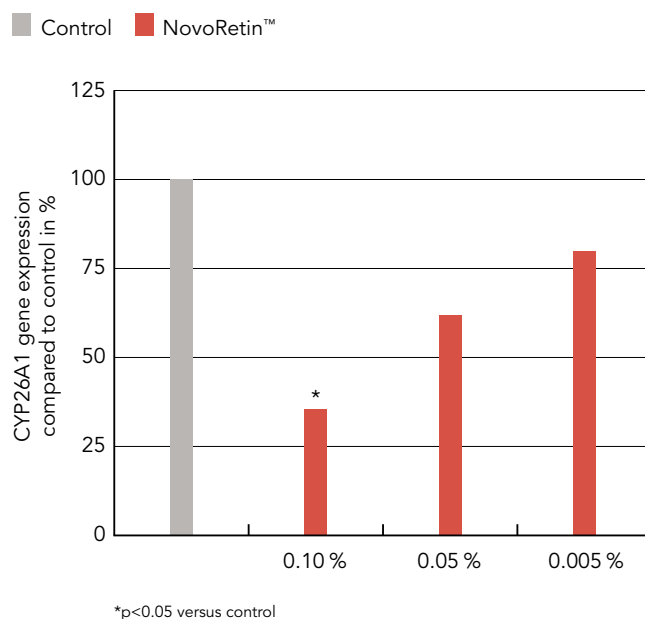
The induction of CYP26A1 expression by retinoic acid was inhibited by NovoRetin™ in a concentration-dependent manner with a maximum inhibition of 65 % at a concentration of 0.1 % NovoRetin™. Therefore, NovoRetin™ inhibits CYP26A1 activity by lowering the expression of this enzyme.

Both experiments indicate that NovoRetin™ has the potential to increase the amount of endogenous retinoic acid in the skin by delaying its rapid degradation. Through this mechanism, NovoRetin™ can exhibit a retinol-like activity.

Inhibition of CYP26A1 Activity



Inhibition of CYP26A1 Gene Expression





Retinoic Acid-Boosting Effect in a 3D Epidermis

Based on its inhibitory potential on CYP26A1 activity, NovoRetin™ was tested for its capacity to enhance retinoic acid activity in a reconstructed human epidermis (RHE). The cells were grown for 7 days in the presence of either 0.008 % NovoRetin™ or 10 nM of the reference compound talarozole, which is a well-known inhibitor of CYP26 enzymes.

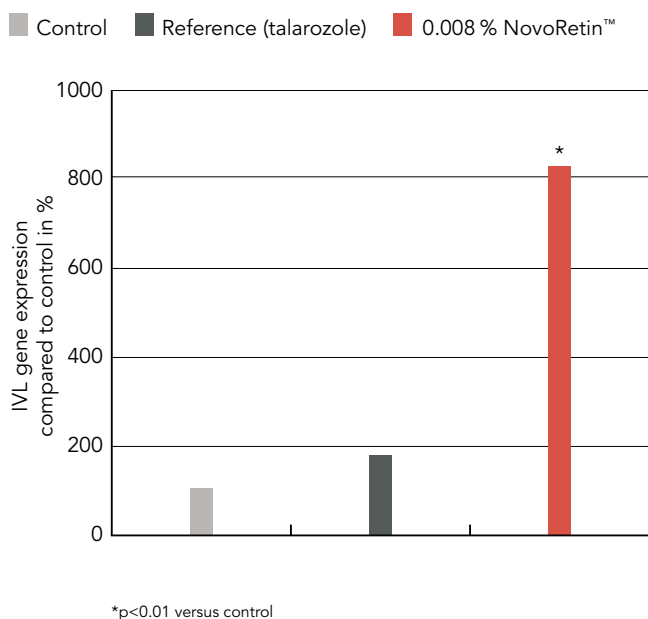
Following this, gene expression of involucrin (IVL), a marker of retinoic acid activity, was determined in each sample.

Results showed that the treatment with retinoic acid led to a slight upregulation of IVL expression in the control. This effect was strongly enhanced by 720 % following the treatment with NovoRetin™, which indicates a high activity of retinoic acid as a result of its inhibited degradation. This effect was even stronger than the upregulation of IVL gene expression induced by the CYP26 inhibitor talarozole.

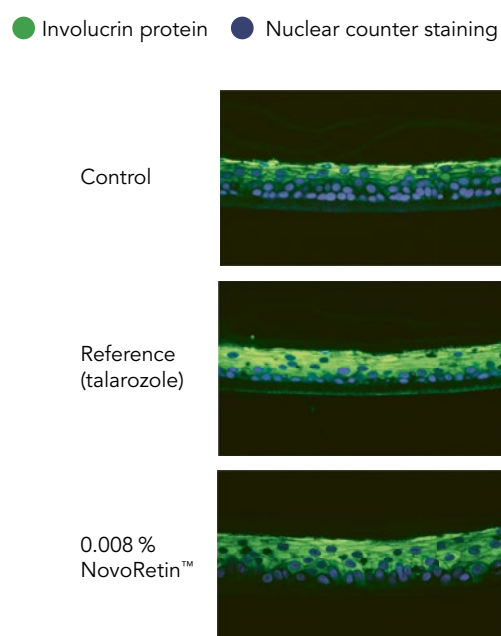
In addition, sections of each sample were used to visualize and quantify involucrin proteins by immunostaining and microscopy. This analysis confirmed the capacity of NovoRetin™ to enhance the effects of retinoic acid. Treatment of the RHE model with NovoRetin™ resulted in a significant upregulation of involucrin protein abundance.

The reinforcement of the retinoic acid-induced effects indicates an enrichment of retinoic acid in the epidermis upon treatment with NovoRetin™. Therefore, a topical application with NovoRetin™ can enhance retinoic acid levels in the skin.

Enhanced Gene Expression of Involucrin



Visualization of Involucrin Proteins



NovoRetin™

Study results



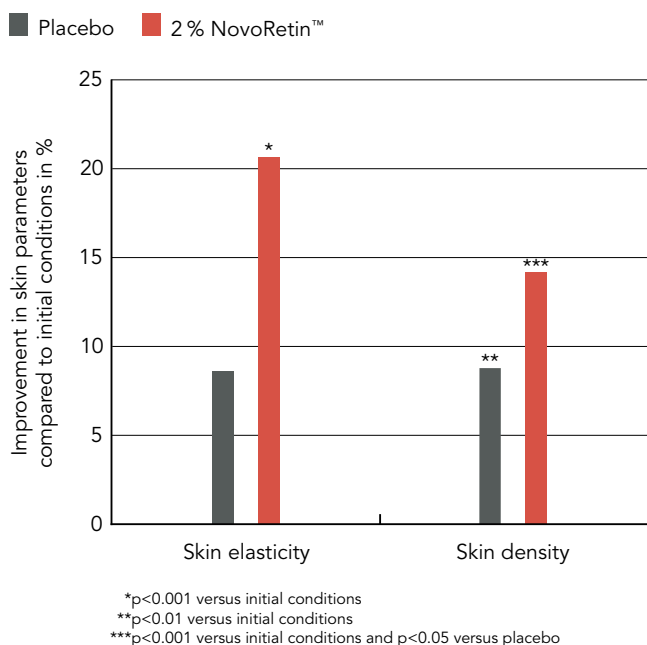
Increased Skin Density and Elasticity

The anti-aging effects of NovoRetin™ were investigated in a placebo-controlled clinical study. For this study, 18 female volunteers aged between 42 and 70 years (mean age: 57 years) with wrinkles in the crow's feet area and signs of photoaging applied a cream with either 2% NovoRetin™ or a corresponding placebo on each side of their face and on different zones of their forearms twice daily for 28 days. The measured parameters at the beginning and at the end of the study were skin elasticity in the face and epidermis-dermis density on the forearm, which were assessed by a cutometer and a Dermalab® Ultrasound device, respectively.

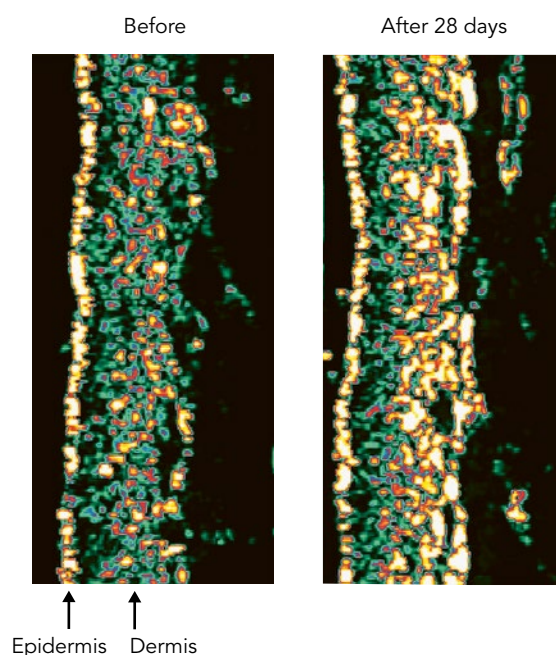
After 28 days of treatment with 2% NovoRetin™, a strong improvement in facial skin elasticity of 20.4% compared to initial conditions was observed. In addition, skin density, measured on the forearm, improved significantly by 13.8%, which was also visible in the ultrasonographic images.

Therefore, by retaining endogenous retinoic acid levels in the skin, NovoRetin™ demonstrates strong anti-aging effects.

Increased Skin Elasticity and Density



Improvement in Skin Density





Immediate Lifting Effect

An additional placebo-controlled clinical study was performed to investigate the immediate anti-wrinkle efficacy of NovoRetin™. For this study, 22 female volunteers aged between 42 and 65 years (mean age: 56 years) with wrinkles in the crow's feet area applied either a cream containing 2% NovoRetin™ or a placebo in the zones around their eyes on each side of their face. Prior to the treatment and 30 minutes after a single application, wrinkle length and depth as well as skin roughness were determined by image analysis of 3D images acquired with a Bio3D Structured-light Scanner and compared to initial conditions. In addition, 2D images were acquired and the volunteers filled in a questionnaire.

NovoRetin™ was able to significantly improve:

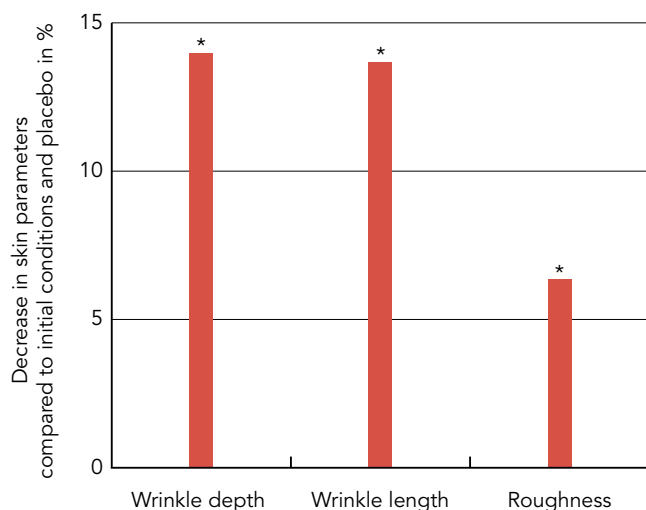
- wrinkle depth by 14.0%
- wrinkle length by 13.7%
- and skin roughness by 6.4% compared to initial conditions and the placebo after only a single application.

The wrinkle reduction was also visible in photographs taken of the volunteers before and 30 minutes after the application. In addition, 82% of the volunteers confirmed that the contour of their eyes appeared smoothed and that the eye region was immediately lifted.

Thus, NovoRetin™ not only provides long-term anti-aging effects due to its retinoic acid-boosting efficacy but also offers immediate lifting effects in the crow's feet area after only a single application.

Reduced Wrinkles and Skin Roughness

■ 2% NovoRetin™



*p<0.05 versus initial conditions and placebo

Visible Decrease in Crow's Feet Wrinkles

Before

After 30 minutes



NovoRetin™

Study results



Pore Refining and Pore Size Reduction

Twenty Caucasian women aged between 24 and 61 years (mean age: 43 years) with enlarged pores applied a cream containing 2% NovoRetin™ on one-half of their face and the corresponding placebo on the other half of their face twice daily for a period of 28 days. Silicon imprints from an area with enlarged pores were recorded before and after the treatment. Meanwhile, pore refinement was determined by measuring the roughness of the skin resulting from the dilation of the pores by PRIMOS.

Results showed that NovoRetin™ significantly refined pores:

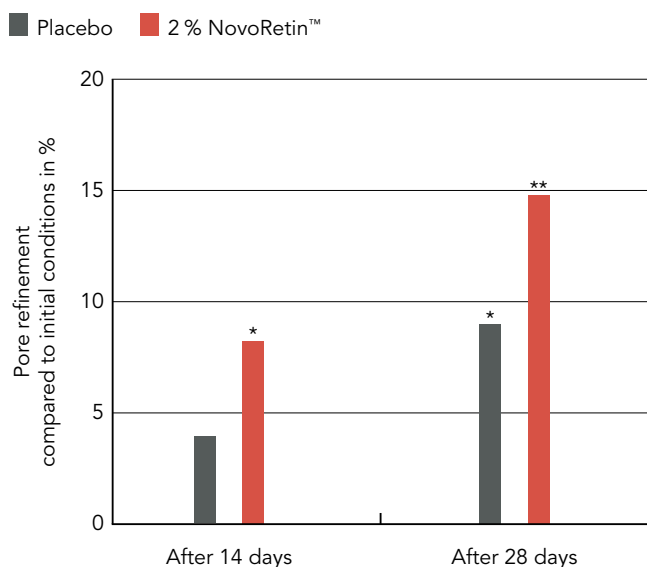
- by more than 8% after 14 days
- by almost 15% after 28 days.

After 28 days of treatment with NovoRetin™, the pore refining effect was observed in 95% of the volunteers.

In the same study, the total pore area was also determined by further calculations of the PRIMOS results. The datasets were filtered to separate point-shaped structures of extreme depth (pores) from the ground structure. Following 28 days of treatment with NovoRetin™, a reduction of the total pore area was observed in 75% of the volunteers.

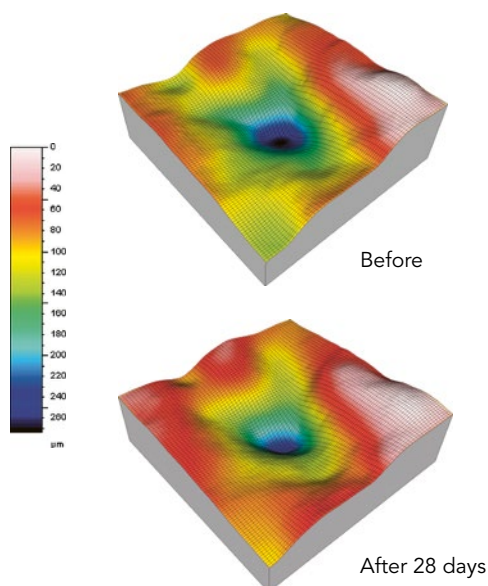
These results demonstrate the pore minimizing and refining potential of NovoRetin™.

Pore Refinement



*p<0.05 versus initial conditions
**p<0.05 versus initial conditions and placebo

Reduction of Pore Size





Pore Count and Volume Reduction

The pore refinement and sebum reduction capabilities of NovoRetin™ were evaluated in a placebo-controlled study that was performed in Seoul, South Korea, on volunteers with oily and impure skin. Twenty-three Asian women aged between 21 and 48 years (mean age: 28 years) applied a cream with 2% NovoRetin™ on one-half of their face and the corresponding placebo on the other half of their face twice daily for a period of 14 days. Pore count and volume were measured on the cheeks of the volunteers with ANTERA 3D.

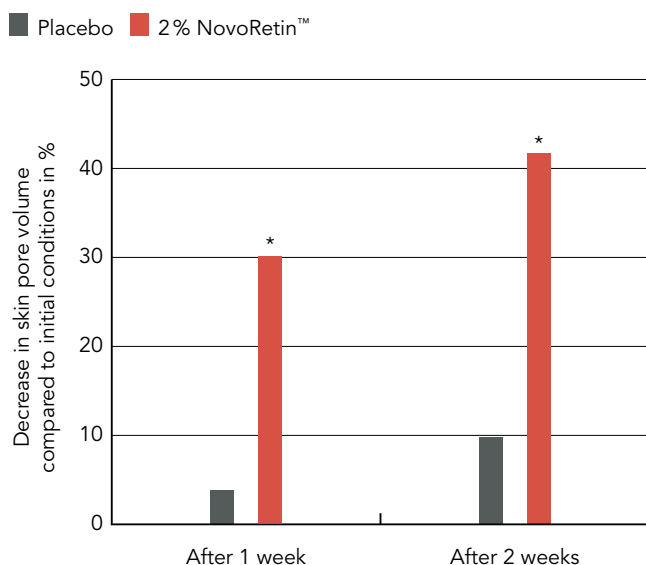
Treatment with 2% NovoRetin™ significantly reduced both pore count and pore volume after just one week by 20.5% and 30.1%, respectively. The effect was even more pronounced after two weeks of treatment with 2% NovoRetin™, whereupon there was a significant reduction of pore count by 30.3% and pore volume by 41.7%.

Visible Reduction of Pore Appearance

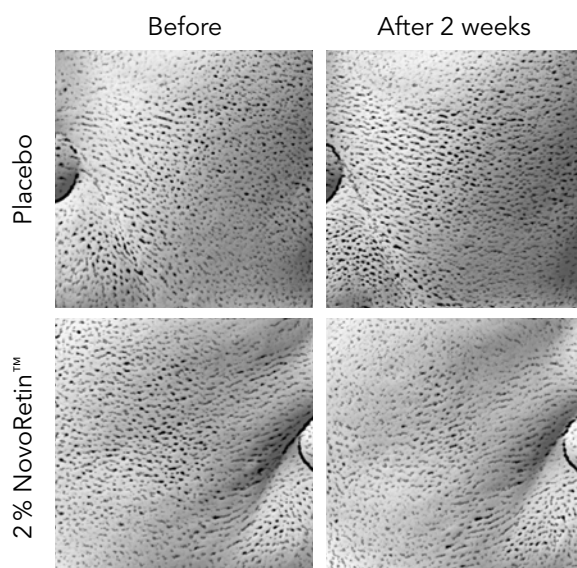
In the same study, the effect of treatment with 2% NovoRetin™ on the appearance of pores was visible in the ANTERA 3D pictures. While the placebo side of the volunteer did not show any improvement in regard to pore appearance, the side of the face that was treated with 2% NovoRetin™ displayed a visible reduction in pore appearance.

These results show an effective reduction in the appearance of pores and an associated improved skin texture after just two weeks of using NovoRetin™.

Decrease in Skin Pore Volume



Reduction of Pore Appearance



The placebo and NovoRetin™ pictures were taken from both halves of the face of the same volunteer.

NovoRetin™

Study results



Anti-Comedogenic Effect

The anti-comedogenic and purifying effects of NovoRetin™ were evaluated in a placebo-controlled study performed in Bangkok, Thailand, on volunteers with acne-prone skin. A total of 44 Thai women aged between 30 and 52 years (mean age: 38 years) with retentional lesions (blackheads and microcysts) on their faces were selected. The volunteers were separated into two groups. The first group applied a cream with 2% NovoRetin™ on their faces twice daily for 28 days, while the second group used a corresponding placebo cream. The comedones (which are also known as blackheads) and microcysts on the faces of the volunteers were counted by a dermatologist before and after the 28-day period of treatment with the test products.

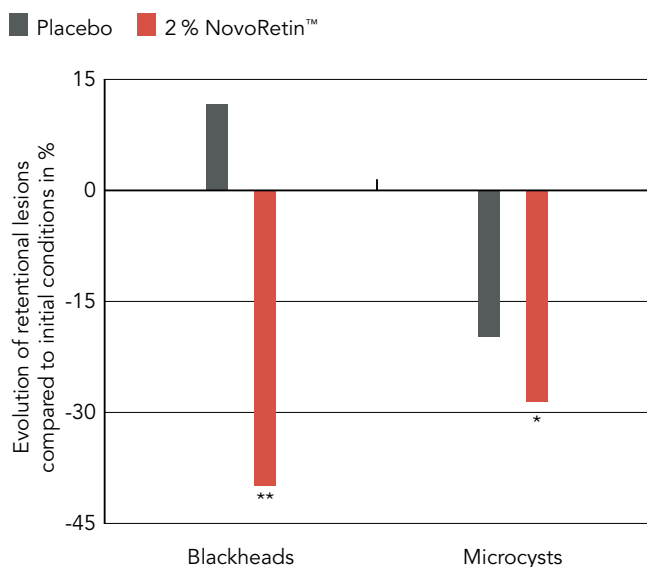
The results showed that NovoRetin™ significantly reduced both types of retentional lesions: blackheads were reduced by almost 40%, while microcysts were reduced by almost 30% compared to initial conditions. NovoRetin™, therefore, exhibits an excellent purifying activity in acne-prone skin.

Visible Reduction of Shininess and Imperfections

In the same study, the shiny aspect of the skin of the volunteers was scored visually by trained assessors both before and at the end of the treatment. A relative reduction of shininess was observed for 32% of the volunteers who were treated with NovoRetin™, whereas the volunteers who were treated with the placebo did not show any visible net reduction in shininess. In addition, macrophotographs were taken both before and after the treatment. An analysis of the pictures of the volunteers treated with the 2% NovoRetin™ emulsion showed a visible reduction of imperfections as well as a reduction in the shininess of the skin.

NovoRetin™ can therefore visibly eliminate shininess and this means it has a matifying effect.

Decrease in Retentional Lesions



Skin Clarifying and Matifying Effect

Before



After





Reduction of Sebum Output

The sebum reduction abilities of NovoRetin™ were also assessed in a placebo-controlled clinical study with 23 Asian women aged between 21 and 48 years (mean age: 28 years), who applied a cream with 2 % NovoRetin™ on one-half of their face and the corresponding placebo on the other half of their face twice daily for a period of 14 days. Sebum production was measured using a sebumeter.

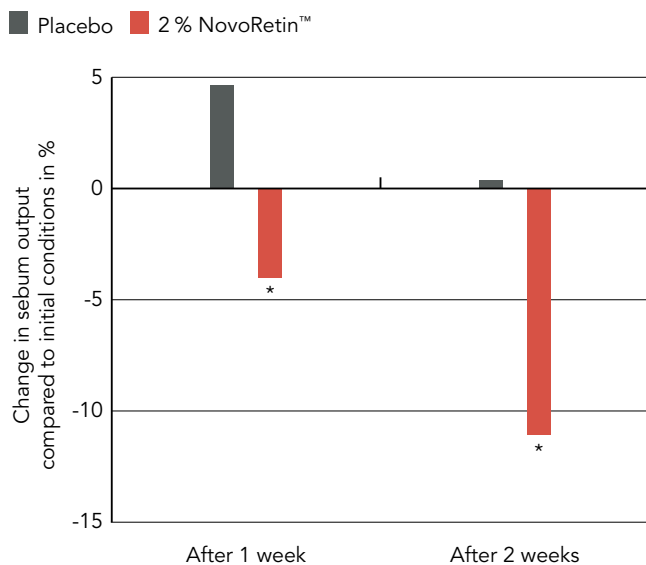
Treatment with 2 % NovoRetin™ reduced sebum output by 4 % after one week and 11 % after two weeks of application. The results were statistically significant compared to the placebo.

Increase in Skin Moisture

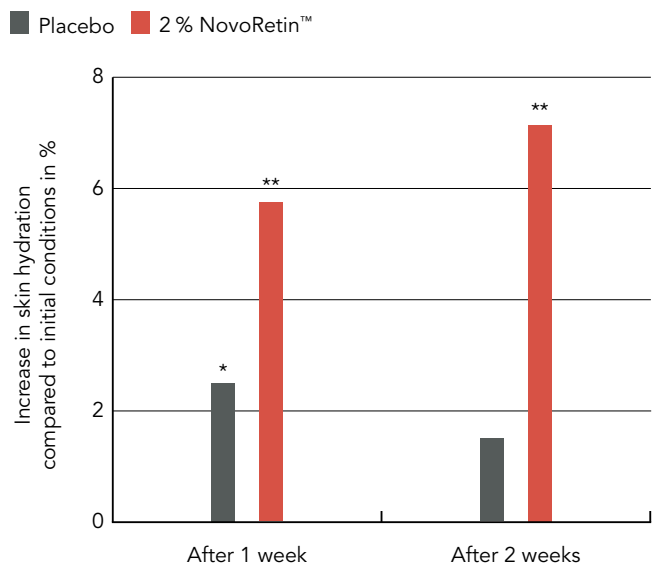
In the same study, skin hydration was measured using a corneometer. This showed that skin hydration significantly increased after treatment with 2 % NovoRetin™.

These results demonstrate that despite efficiently reducing sebum production in oily skin types and in contrast to classic retinol applications, treatment with NovoRetin™ does not dry out the skin. On the contrary, skin hydration is even promoted.

Decrease in Sebum Output



Increase in Skin Hydration



NovoRetin™

The natural alternative to retinol

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- Boosts naturally occurring retinoic acid in the skin
- Improves skin elasticity and density
- Effects immediate lifting
- Visibly refines pores and blemishes
- Stable and safe retinol activity

Applications

- Advanced age-defying formulas
- Immediate lifting products
- Treatments for acne-prone skin
- Pore minimizing formulas

Marketing Benefits

- A plant-based and gentle retinol alternative
- Very special and endemic source from the Mediterranean region
- All-in-one: increases skin density and elasticity, reduces pore size and wrinkles, refines skin texture, and creates a smooth matte finish
- Gentle yet highly effective – without the risk of increased photosensitivity and drying out the skin
- 100% natural origin

Innovating for your success

Mibelle Biochemistry designs and develops innovative, high-quality actives based on naturally derived compounds and profound scientific know-how. Inspired by nature – Realized by science.



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